Schur, WSC, and c_0 Classes for Almost Multi-Boundedness

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Source and Target

The source paper is Hung Le Pham, *A combinatorial characterisation of amenable locally compact groups*, arXiv:2310.19178. Problem 4.8 asks:

> Determine those classes of Banach spaces in which relative weak compactness implies and/or is implied by almost (p, q)-multi-boundedness.

The paper proves the equivalence for $\mathfrak{L}^1$-spaces and gives examples showing that neither implication holds in arbitrary Banach spaces. The result below gives additional positive and negative classes for every $1 \le p, q < \infty$.

Definitions

For a Banach space E, $x = (x_1, \dots, x_n) \in E^n$, and $1 \le p, q < \infty$, Pham defines

$$\|x\|_n^{(p,q)} = \sup \left\{ \left(\sum_{j=1}^n |\lambda_j(x_j)|^q \right)^{1/q} : \lambda_1, \dots, \lambda_n \in E^*, \sup_{\|u\| \le 1} \left(\sum_{j=1}^n |\lambda_j(u)|^p \right)^{1/p} \le 1 \right\}.$$

For $B \subset E$, write $c_{p,q}(B)$ for the common quantity in Pham's Proposition 3.4:

$$c_{p,q}(B) = \lim_{n \to \infty} \sup_{x_1, \dots, x_n \in B} \frac{\|(x_1, \dots, x_n)\|_n^{(p,q)}}{n^{1/q}}.$$

The set B is almost (p, q)-multi-bounded when $c_{p,q}(B) = 0$.

Proof Intuition

The tuple norms are strong enough to detect separated ℓ_1-structure, but weak enough that finite sets disappear after normalization by $n^{1/q}$. A norm-compact set is approximated by a finite set, so it is automatically almost multi-bounded. For the implication from almost multi-boundedness to weak compactness, Rosenthal's dichotomy is decisive: in weakly sequentially complete spaces a bounded non-weakly compact set must contain an ℓ_1 basic sequence, and such a sequence is seen by the tuple norms. On the other hand, the summing basis in c_0, already isolated in Pham's examples, gives an almost multi-bounded set that is not weakly compact; this obstruction passes to every space containing c_0.

Lemmas

Lemma 1 (Finite sets vanish asymptotically). *Let A be a finite subset of a Banach space E. Then $c_{p,q}(A) = 0$ for every $1 \leq p, q < \infty$.*

Proof. Write $A = \{a_1, \dots, a_m\}$. Let $x_1, \dots, x_n \in A$, and for each r put $I_r = \{j : x_j = a_r\}$. If $\lambda_1, \dots, \lambda_n \in E^*$ satisfy

$$\sup_{\|u\| \leq 1} \left(\sum_{j=1}^{n} |\lambda_j(u)|^p \right)^{1/p} \leq 1,$$

then

$$\left(\sum_{j \in I_r} |\lambda_j(a_r)|^p \right)^{1/p} \leq \|a_r\| .$$

If $q \geq p$, the corresponding q-norm is at most $\|a_r\|$. If $q < p$, it is at most $n^{1/q-1/p} \|a_r\|$. Hence

$$\|(x_1, \dots, x_n)\|_n^{(p,q)} \leq \left(\sum_{r=1}^{m} \|a_r\| \right) n^{\max\{0, 1/q-1/p\}}.$$

After division by $n^{1/q}$, this tends to 0. □

Lemma 2 (Norm-compact sets are almost multi-bounded). *Let B be norm relatively compact in a Banach space E. Then $c_{p,q}(B) = 0$ for every $1 \leq p, q < \infty$.*

Proof. Fix $\eta > 0$, and choose a finite η-net A for B. For $x_1, \dots, x_n \in B$, choose $a_j \in A$ with $\|x_j - a_j\| \leq \eta$. Every admissible tuple (λ_j) in the definition of $\|\cdot\|_n^{(p,q)}$ satisfies $\|\lambda_j\| \leq 1$ for each j. Thus

$$\|(x_1 - a_1, \dots, x_n - a_n)\|_n^{(p,q)} \leq \eta n^{1/q}.$$

By Lemma 1,

$$\limsup_{n \to \infty} \sup_{x_1, \dots, x_n \in B} \frac{\|(x_1, \dots, x_n)\|_n^{(p,q)}}{n^{1/q}} \leq \eta.$$

Letting $\eta \downarrow 0$ proves the claim. □

Lemma 3 (ℓ_1-basic obstruction). *If $B \subset E$ contains a sequence equivalent to the unit vector basis of ℓ_1, then B is not almost (p,q)-multi-bounded for any $1 \leq p, q < \infty$.*

Proof. Pass to a sequence $(x_j) \subset B$ and choose $\kappa > 0$ such that

$$\left\| \sum_{j=1}^{n} \alpha_j x_j \right\| \geq \kappa \sum_{j=1}^{n} |\alpha_j|$$

for all finite scalar families (α_j). The coordinate map $\sum \alpha_j x_j \mapsto (\alpha_j)$ from $[x_j]$ into ℓ_1 has norm at most κ^{-1}. Let $\lambda_j \in E^*$ be Hahn–Banach extensions of its coordinates. For every $u \in E$,

$$\left(\sum_{j=1}^{n} |\lambda_j(u)|^p \right)^{1/p} \leq \sum_{j=1}^{n} |\lambda_j(u)| \leq \kappa^{-1} \|u\| .$$

Therefore $(\kappa\lambda_1, \dots, \kappa\lambda_n)$ is admissible in the definition of the (p,q)-multi-norm, and

$$\|(x_1, \dots, x_n)\|_n^{(p,q)} \geq \left(\sum_{j=1}^{n} |\kappa\lambda_j(x_j)|^q \right)^{1/q} = \kappa n^{1/q}.$$

Hence $c_{p,q}(B) \geq \kappa > 0$. □

Lemma 4 (Functoriality)**.** *Let $T : E \to F$ be a bounded linear operator. If $B \subset E$ is almost (p,q)-multi-bounded, then $T(B) \subset F$ is almost (p,q)-multi-bounded.*

Proof. For every $x_1, \dots, x_n \in E$ and every admissible tuple $(\lambda_1, \dots, \lambda_n) \in (F^*)^n$, the tuple $(T^*\lambda_1, \dots, T^*\lambda_n)$ has weak p-summing norm at most $\|T\|$. Therefore

$$\|(Tx_1, \dots, Tx_n)\|_n^{(p,q)} \leq \|T\| \; \|(x_1, \dots, x_n)\|_n^{(p,q)}.$$

After taking suprema and normalizing by $n^{1/q}$, the assertion follows. □

Theorem

Theorem 1. *Let $1 \leq p, q < \infty$.*

(a) *In every Banach space, norm relatively compact subsets are almost (p,q)-multi-bounded.*

(b) *If E has the Schur property, then a subset $B \subset E$ is relatively weakly compact if and only if it is almost (p,q)-multi-bounded.*

(c) *If E is weakly sequentially complete, then every almost (p,q)-multi-bounded subset of E is relatively weakly compact.*

(d) *If E contains an isomorphic copy of c_0, then the implication "almost (p,q)-multi-bounded implies relatively weakly compact" fails in E.*

Proof. Part (a) is Lemma 2.

Assume now that E has the Schur property. If B is relatively weakly compact, then B is norm relatively compact by Eberlein–Smulian and the Schur property; part (a) gives almost (p,q)-multi-boundedness.

Conversely, suppose B is almost (p,q)-multi-bounded. Pham observes after Definition 3.5 that unbounded sets have $c_{p,q}(B) = \infty$, so B is bounded. Since Schur spaces are weakly sequentially complete, the argument in the next paragraph shows that B is relatively weakly compact. This proves the equivalence in Schur spaces.

Assume more generally that E is weakly sequentially complete, and let B be almost (p,q)-multi-bounded. Again B is bounded. If B were not relatively weakly compact, Eberlein–Smulian would give a sequence in B with no weakly convergent subsequence. Rosenthal's ℓ_1 theorem gives either an ℓ_1-basic subsequence or a weakly Cauchy subsequence. The first possibility contradicts Lemma 3. The second contradicts weak sequential completeness. Hence B is relatively weakly compact.

Finally suppose that E contains a closed subspace Y isomorphic to c_0. In the examples following Theorem 4.4, Pham records that the summing basis

$$s_n = e_1 + \cdots + e_n \qquad (n \in \mathbb{N})$$

forms a subset $S = \{s_n : n \in \mathbb{N}\}$ of c_0 that is almost (p, q)-multi-bounded for every p, q, but is not relatively weakly compact. If $J : c_0 \to Y \subset E$ is an isomorphism, Lemma 4 shows that $J(S)$ is almost (p, q)-multi-bounded in E. It is not relatively weakly compact: weak compactness of $J(S)$ in E would imply weak compactness in the closed subspace Y, and then weak compactness of S in c_0 after applying J^{-1}. $\square$

Corollary 1. *Every finite-dimensional Banach space has the full equivalence:*

$$B \textit{ is relatively weakly compact} \iff B \textit{ is almost } (p, q)\textit{-multi-bounded.}$$

Proof. Finite-dimensional spaces are reflexive, hence weakly sequentially complete, and have the Schur property. $\square$

Corollary 2. *Every reflexive Banach space satisfies the implication from almost (p, q)-multi-boundedness to relative weak compactness.*

Proof. Reflexive spaces are weakly sequentially complete. $\square$

Verification Notes and Limitations

This is a partial answer to Problem 4.8, not a classification. It adds exact families to the landscape: Schur spaces for the two-sided equivalence, weakly sequentially complete spaces for the almost-implies-weak-compactness direction, and spaces containing c_0 as a broad negative class for that same direction. It is consistent with Pham's examples: infinite-dimensional reflexive unit balls can fail to be almost multi-bounded for some parameters, so the reverse implication does not follow from reflexivity alone. The remaining frontier for the almost-implies-weak-compactness direction includes the classical non-weakly-sequentially-complete spaces that do not contain c_0, where Rosenthal's strongly summing alternative can occur.

Bounded novelty search on 2026-06-20 checked exact phrases around almost (p, q)-multi-boundedness, Schur spaces, weak sequential completeness, c_0, weak compactness, $c_{p,q}(B)$, and arXiv:2310.19178. No exact existing Schur/WSC/c_0 statement was found. The result should be reviewed as a partial landscape theorem rather than as a full solution of the source problem.

References

References

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